WORKSHEET 6.11

Choosing an Integration Method

Unit 6: Integration and Accumulation of Change

Key ideas

- First ask whether **algebra** is enough: expand a product, or divide a quotient term by term.
- Then look for a **substitution**: an inside function whose derivative is present, up to a constant.
- A numerator that is the derivative of the denominator gives a **logarithm**.
- A constant over a quadratic with no real zeros gives an **arctangent**, after completing the square.

Worked examples

Example 1 · A logarithm

Find $\displaystyle\int \frac{2x+1}{x^2+x+3}\,dx$.

Solution. The numerator is the derivative of the denominator.

$\ln(x^2+x+3)+C$.

Example 2 · Algebra first

Find $\displaystyle\int \frac{(x+1)^2}{\sqrt{x}}\,dx$.

Solution. Expand and divide: $x^{3/2}+2x^{1/2}+x^{-1/2}$.

$\frac{2}{5}x^{5/2}+\frac{4}{3}x^{3/2}+2\sqrt{x}+C$.

Example 3 · An arctangent

Find $\displaystyle\int \frac{1}{x^2+2x+2}\,dx$.

Solution. $x^2+2x+2=(x+1)^2+1$.

$\arctan(x+1)+C$.

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Practice

CORE

1. Find $\int \left(3x^2+1\right)\left(x^3+x\right)^5\,dx$.

2. Find $\int \dfrac{x^2+1}{x^2}\,dx$.

3. Find $\int \dfrac{1}{1+4x^2}\,dx$.

4. Find $\int \dfrac{e^x}{e^x+2}\,dx$.

EXAM LEVEL

5. Find $\int \dfrac{\sin x}{\cos^2 x}\,dx$.

6. Find $\int \dfrac{x}{x+3}\,dx$.

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7. Find $\displaystyle\int \frac{e^{2x}}{1+e^{4x}}\,dx$.

8. Find $\displaystyle\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

Question 9 · Free-response practice

Evaluate each integral. Name the method you use.

(a) $\displaystyle\int_0^1 \frac{x+2}{x^2+4x+1}\,dx$

(b) $\displaystyle\int_0^1 \frac{x}{x+1}\,dx$

(c) $\displaystyle\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\,dx$

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Worked solutions

1. Substitution $u = x^3 + x$: $\dfrac{(x^3+x)^6}{6} + C$.

2. Divide: $\displaystyle\int \left(1 + x^{-2}\right) dx = x - \frac{1}{x} + C$.

3. Arctangent with $u = 2x$: $\dfrac{1}{2}\arctan(2x) + C$.

4. Substitution $u = e^x + 2$: $\ln(e^x + 2) + C$.

5. $u = \cos x$: $\displaystyle\int -u^{-2}\, du = \frac{1}{u} = \sec x + C$.

6. Division: $1 - \dfrac{3}{x+3}$, giving $x - 3\ln|x+3| + C$.

7. $u = e^{2x}$, $du = 2e^{2x}dx$: $\dfrac{1}{2}\arctan\left(e^{2x}\right) + C$.

8. $u = \ln x$: $\sin(\ln x) + C$.

9. (a) Substitution $u = x^2 + 4x + 1$, $du = 2(x+2)\,dx$, limits 1 to 6: $\dfrac{1}{2}\ln 6$.

   (b) Division: $\displaystyle\int_0^1 \left(1 - \frac{1}{x+1}\right) dx = 1 - \ln 2$.

   (c) Inverse sine: $\Big[\arcsin x\Big]_0^{1/2} = \dfrac{\pi}{6}$.

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